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Convolution Semigroups and Resolvent Families of Measures on Hypergroups

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1. Introduction

This is a continuation of studies on measures on a hypergroup begun by the authors in [3] and [4]. We commence the discussion in Sect. 2 with the definitions of strongly positive definite and strongly negative definite functions on the hypergroup dual $K^\wedge$ of a hypergroup K. Strong positive definiteness axiomatizes Bochner's theorem for positive definite functions on $K^\wedge$ in terms of representing measures on K, and strong negative definiteness axiomatizes Schoenberg's duality result for negative definite functions on $K^\wedge$ in terms of the positive definiteness of their exponentials. The properties of the set of all strongly negative definite functions on $K^\wedge$ established in Sect. 2 are applied, in Sect. 3, to obtain an analogue of the classical Schoenberg correspondence to hypergroups, a result which ensures that to every continuous convolution semigroup of positive contraction measures on K there corresponds a unique strongly negative definite function on $K^\wedge$. Sections 4 and 5 are devoted to the bijective correspondence between continuous convolution semigroups of contraction measures and resolvent families on K. The analysis includes useful facts about the correspondence of all supports of both kinds of objects and results on the periodicity of strongly negative definite functions on $K^\wedge$. Section 6 on positive definite measures on a hypergroup is of a preparative nature for the main applications on transience of convolution semigroups to be treated in Sect. 7. Positive definite measures that are not necessarily bounded play an important role in the potential theory of continuous convolution semigroups. We take up this part of the theory in order to study the potential measures corresponding to such semigroups. The transience criterion (see Theorems 7.9 and 7.10) requires refined investigation on the local integrability of the inverse of the strongly negative definite function corresponding to the semigroup.

Concerning the methods applied in the present work we have been inspired and lead all the way through by the classical results available in the case of a locally compact abelian group, which are treated in some detail in the monograph [2] by Berg and Forst. However it has been necessary to take a different approach to the proofs within the framework of hypergroups, because of various technical problems that have to be overcome; one of the more fundamental of these is that, in general, hypergroups lack a genuine Pontrjagin duality.

For the entire presentation we adopt the notion of a hypergroup K in the sense of Jewett [7]. We are mainly concerned with commutative hypergroups which, by a deep result of Spector [13], admit a Haar measure ω_K. Moreover we assume throughout that $K^\wedge$ is also a hypergroup with respect to pointwise multiplication. In this case we can fix Haar measures ω_K, $\omega_{K^\wedge}$ on $K, K^\wedge$ respectively such that Plancherel's theorem holds with Plancherel measure π equal to $\omega_{K^\wedge}$, and then $\mathrm{supp}(\pi)=\mathrm{supp}(\omega_{K^\wedge})=K^\wedge$; see [7], Sect. 12.4 and [6], Sect. 3.

If $K^\wedge$ is a hypergroup under pointwise multiplication then there is a natural embedding $K\subset K^{\wedge\wedge}$. In case equality holds here we refer to K as being a *strong hypergroup*. Examples of strong hypergroups are given in [6], Sect. 3.8; they include the Jacobi hypergroups $\mathbb{Z}^+$ and their duals, orbit hypergroups of the form G_B for a locally compact group G and a relatively compact subgroup B of $\mathrm{Aut}(G)$, and their duals, and in particular certain types of conjugacy hypergroups and the orbit hypergroup corresponding to Euclidean motion. It should be noted that the ultraspherical hypergroups, which are special Jacobi hypergroups, appear as duals in the harmonic analysis of homogeneous spaces (see [6], Sect. 1.3.2).

To complete this introduction we clarify some notation (consistent with [3] and [4]). If K is a hypergroup then $x\to x^-$ denotes its involution. $C(K)$, $C^b(K)$, $C_0(K)$ and $C_c(K)$ will be used for the spaces of all continuous functions on K, those that are bounded, those that vanish at infinity, and those with compact support respectively. By $M(K)$, $M^b(K)$, $M^b_+(K)$ and $M^1(K)$ we denote all Radon measures, the bounded ones, those that are bounded and nonnegative, and the probability measures on K respectively. For any $f\in C(K)$ and $\mu\in M(K)$, f^- and μ^- are defined through involution, and $f^\sim$ and $\mu^\sim$ through involution and complex conjugation at the same time. For a given subset A of K, A^c stands for the closure of A and ξ_A for the indicator function of A. ε_x denotes the Dirac measure at $x\in K$. The unit character of K (which becomes the distinguished unit element of $K^\wedge$ if $K^\wedge$ is a hypergroup) is written as $\mathbb{1}$. Finally, if K is compact then ω_K will be normalised to have unit total variation.

2. Strongly Negative Definite Functions

Let K be a commutative hypergroup such that its dual $K^\wedge$ is a hypergroup under pointwise multiplication.

2.1. **Definition.** Let Λ be a nonempty subset of $K^\wedge$. A function $\phi\in C(\Lambda)$ is called *strongly positive definite* on Λ if there exists a measure $\mu\in M^b_+(K)$ (which

is necessarily unique) satisfying

$$\hat{\mu}(\chi)=\begin{cases}\phi(\chi) & \text{for } \chi\in\Lambda,\\ 0 & \text{for } \chi\notin\Lambda.\end{cases}$$

A function $\psi\in C(\Lambda)$ is said to be *strongly negative definite* if $\psi(\mathbb{1})\geqq 0$ and $\exp(-t\psi)$ is strongly positive definite on Λ for each $t>0$.

The sets of strongly positive definite and strongly negative definite functions on Λ will be abbreviated by $SP(\Lambda)$ and $SN(\Lambda)$ respectively; when $\Lambda=K^\wedge$ we omit reference to the support set.

2.2. **Discussion.** Clearly each strongly positive definite function is positive definite in the sense of [3], Sect. 3.1, but the converse implication does not hold. An equivalent definition of strong positive definiteness (on $K^\wedge$) has been given by Lasser in [9], who shows (Proposition 4) that for strong hypergroups K $(=K^{\wedge\wedge})$ strong positive definiteness is equivalent to (bounded) positive definiteness. The existence of a hypergroup K (with dual a hypergroup under pointwise multiplication) such that $K\subsetneqq K^{\wedge\wedge}$ implies the existence of a (bounded) positive definite function on $K^\wedge$ that is not strongly positive definite.

We also note that each strongly negative definite function is negative definite in the sense of [8], Sect. 3.1; see Theorem 3.1.5 of [8] which is an analogue of one half of Schoenberg's duality result. It is not known for which hypergroups, negative definiteness implies strong negative definiteness (compare [11], Sect. 2).

In view of the above discussion we obtain the following.

2.3. **Proposition.** *Let* $\psi\in SN(K^\wedge)$. *Then*

(a) $c\mathbb{1}\in SN(K^\wedge)$ *for all* $c\geqq 0$.

(b) $\psi(\chi*\chi^-)\in\mathbb{R}$ *for all* $\chi\in K^\wedge$.

(c) $\psi(\chi^-)=\overline{\psi(\chi)}$ *for all* $\chi\in K^\wedge$.

(d) $\psi(\chi)+\psi(\chi^-)\geqq\psi(\chi*\chi^-)$ *for all* $\chi\in K^\wedge$.

(e) *If* ψ *is bounded or satisfies* $\operatorname{Re}\psi\geqq 0$ *then* $\operatorname{Re}\psi\geqq\psi(\mathbb{1})\geqq 0$.

(f) *If* $\operatorname{Re}\psi\geqq 0$ *then*

$$\psi(\chi_1*\chi_2)^{\frac{1}{2}}\leqq|\psi(\chi_1)|^{\frac{1}{2}}+|\psi(\chi_2)|^{\frac{1}{2}}$$

whenever $\chi_1,\chi_2\in K^\wedge$.

The proofs of the preceding statements, which parallel those in the classical case, have been given in [8].

2.4. **Proposition.** *If* $\phi\in SP(K^\wedge)$ *then* $\psi:=\phi(\mathbb{1})-\phi\in SN(K^\wedge)$.

Proof. Clearly $\psi\in C(K^\wedge)$ and $\psi(\mathbb{1})\geqq 0$. We have to show that $\exp(-t\psi)\in SP(K^\wedge)$ for all $t>0$.

Consider $t>0$ fixed and, for each $n\geqq 1$, write

$$\phi_n:=\left(1-\frac{t(\phi(\mathbb{1})-\phi)}{n}\right)^n.$$

By assumption there exists a measure $\mu \in M_+^b(K)$ such that $\hat{\mu} = \phi$. For every $n \geqq 1$ we introduce the measure

$$\nu_n := \left(\varepsilon_e - \frac{t(\|\mu\| \varepsilon_e - \mu)}{n}\right)^{*n} \in M^b(K),$$

where $*n$ denotes the n-fold convolution of the term in the main brackets. It turns out that for n sufficiently large ($\geqq t\|\mu\|$) the measures

$$\nu_n' := \varepsilon_e - \frac{t(\|\mu\| \varepsilon_e - \mu)}{n}$$

belong to $M^+(K)$. Moreover

$$\lim_{n\to\infty} \hat{\nu}_n(\chi) = \lim_{n\to\infty} \left(1 - \frac{t(\phi(\mathbb{1}) - \phi(\chi))}{n}\right)^n = \exp[-t(\phi(\mathbb{1}) - \phi(\chi))]$$

for all $\chi \in K^\wedge$. From Lévy's continuity theorem ([3], Theorem 4.6) we then infer that there exists a measure $\nu \in M_+^b(K)$ with

$$\hat{\nu} = \exp[-t(\phi(\mathbb{1}) - \phi)],$$

whence $\psi \in SN(K^\wedge)$.

2.5. **Proposition.** *Let* $\psi \in SN(K^\wedge)$ *and* $\psi(\mathbb{1}) > 0$. *Then* $\dfrac{1}{\psi} \in SP(K^\wedge)$.

Proof. By assumption $\exp(-t\psi) \in SP(K^\wedge)$ for all $t > 0$. Thus for all $t > 0$, $\chi \in K^\wedge$ we obtain

$$|\exp(-t\psi)| \leqq \exp(-t\psi(\mathbb{1})),$$

and it follows that $\psi(\chi) > 0$ as well as

$$\frac{1}{\psi(\chi)} = \int_0^\infty \exp(-t\psi(\chi))\, dt = \int_0^\infty \hat{\mu}_t(\chi)\, dt,$$

where μ_t is the representing measure for $\exp(-t\psi)$. Furthermore

$$\int_0^\infty \hat{\mu}_t(\chi)\, dt = \int_0^\infty \mu_t(\bar{\chi})\, dt = \nu(\chi) = \hat{\nu}(\chi),$$

where

$$\nu := \int_0^\infty \mu_t\, dt \geqq 0.$$

Finally

$$\nu(\mathbb{1}) = \frac{1}{\psi(\mathbb{1})} < \infty,$$

and consequently $\nu \in M_+^b(K)$, thus proving the assertion.

3. The Schoenberg Correspondence

We start by studying in more detail properties of the annihilator, defined for a nonempty subset H of an arbitrary commutative hypergroup K by

$$A(K^\wedge, H) := \{\chi \in K^\wedge : \chi(x) = 1 \text{ for all } x \in H\},$$

and for a nonempty subset Λ of $K^\wedge$ by

$$A(K, \Lambda) := \{x \in K : \chi(x) = 1 \text{ for all } \chi \in \Lambda\}.$$

3.1. **Proposition.** *Let K be a commutative hypergroup. Then*

(a) $A(K^\wedge, H) = A(K^\wedge, [H])$, *where $[H]$ denotes the smallest subhypergroup of K containing H.*

(b) $A(K^\wedge, \{e\}) = K^\wedge$.

(c) $A(K^\wedge, H) \neq K^\wedge$ *if* $H \neq \{e\}$.

(d) $A(K^\wedge, H)$ *is closed in* $K^\wedge$.

(e) *If H is a compact subhypergroup of K then $A(K^\wedge, H)$ is open in $K^\wedge$.*

(f) *If H is not locally ω_K-null then $A(K^\wedge, H)$ is compact.*

(g) $A(K, \Lambda)$ *is a subhypergroup of* K.

(h) *If Λ is not locally $\omega_{K^\wedge}$-null then $A(K, \Lambda)$ is compact.*

If $K^\wedge$ is a hypergroup under pointwise multiplication then we also have

(i) $A(K^\wedge, H)$ *is a subhypergroup of* $K^\wedge$.

Proof. (a) Since $H \subset [H]$, one inclusion is clear. In the other direction we note that $A(K^\wedge, H) = A(K^\wedge, H^-)$ and

$$[H] = \bigcup \{x_1 * \ldots * x_n : x_i \in H \cup H^- \text{ for all } i = 1, \ldots, n; n \in \mathbb{N}\}.$$

We follow an induction argument. If $\chi(x_1) = \chi(x_2) = 1$ then

$$1 = \chi(x_1)\chi(x_2) = \chi(x_1 * x_2) = \int_K \chi \, d(\varepsilon_{x_1} * \varepsilon_{x_2}).$$

But we have $\varepsilon_{x_1} * \varepsilon_{x_2} \in M^1(K)$ and, by [7], Lemma 6.3D, $\|\chi\|_\infty = 1$. Therefore $\chi = 1$ on $\operatorname{supp}(\varepsilon_{x_1} * \varepsilon_{x_2}) = \{x_1\} * \{x_2\}$. Assume that $\chi = 1$ on $\{x_1\} * \ldots * \{x_m\}$ and $\chi(x_{m+1}) = 1$, and choose $x \in \{x_1\} * \ldots * \{x_m\} * \{x_{m+1}\}$. Then by [7], Lemma 3.2E, we obtain $x \in \{y\} * \{x_{m+1}\}$ for some $y \in \{x_1\} * \ldots * \{x_m\}$, and the assumption, together with what has been shown previously, yields $\chi(x) = 1$.

(b) This statement is clear.

(c) To prove this just use the fact that $K^\wedge$ separates the points of K (see [14], Theorem 2.4).

(d) If $H \neq \{e\}$ then by (c) choose $\chi_0 \notin A(K^\wedge, H)$ and $a \in H$ with $\chi_0(a) \neq 1$. Then

$$\{\chi \in K^\wedge : |\chi(a) - \chi_0(a)| < |\chi_0(a) - 1|\} \subset A(K^\wedge, H)^c,$$

and the first set is a neighbourhood of χ_0.

(e) Choose $\delta \in (0, 1)$ and write

$$P(H, \delta) := \{\chi \in K^\wedge : |\chi(x) - 1| < \delta \text{ for all } x \in H\}.$$

Clearly $A(K^\wedge, H) \subset P(H, \delta)$, where $P(H, \delta)$ is an open set since H is compact. We assume that there exists $\chi \in P(H, \delta) \setminus A(K^\wedge, H)$ and write

$$\varepsilon := \max_{x \in H} |\chi(x) - 1| := |\chi(z) - 1|,$$

for suitable $z \in H$. By assumption $0 < \varepsilon < \delta$. Then

$$\begin{aligned} \varepsilon |\chi(z) + 1| = |\chi(z)^2 - 1| = |\chi(z * z) - 1| &= |\int_K (\chi - 1)\, d(\varepsilon_z * \varepsilon_z)| \\ &\leqq \int_K |\chi - 1|\, d(\varepsilon_z * \varepsilon_z) \\ &\leqq \varepsilon \end{aligned}$$

since $\{z\} * \{z\} \subset H$, and hence $|\chi(z) + 1| \leqq 1$. However

$$|\chi(z) - 1| = \varepsilon < 1$$

and we obtain a contradiction. Thus no such χ exists and $A(K^\wedge, H) = P(H, \delta)$ is open in $K^\wedge$.

(f) First we note that for any compact subset N of K with $\omega_K(N) > 0$ and any $\delta \in (0, 1)$ the open set $P(N, \delta)$ is relatively compact. The proof of this statement, which is similar to that of [5], Corollary (23.16), uses the Riemann-Lebesgue lemma for hypergroups (see [7], Theorem 7.3 H.) Now let H be a subset of K that is not locally ω_K-null, so that $\omega_K(N) > 0$ for some compact subset N of H. Then applying the above we obtain a relatively compact open subset $P(N, \delta)$ containing $A(K^\wedge, H)$. Thus, using (d), $A(K^\wedge, H)$ is compact.

(g) Clearly $e \in A(K, \Lambda)$ and $A(K, \Lambda)^- = A(K, \Lambda)$. Consider $x_1, x_2 \in A(K, \Lambda)$. Then for all $\gamma \in \Lambda$,

$$\int_K \overline{\gamma}\, d\varepsilon_{x_1} * \varepsilon_{x_2} = \hat{\varepsilon}_{x_1}(\gamma)\, \hat{\varepsilon}_{x_2}(\gamma) = \overline{\gamma}(x_1)\, \overline{\gamma}(x_2) = 1.$$

Since $|\overline{\gamma}| \leqq 1$ and $\varepsilon_{x_1} * \varepsilon_{x_2} \in M^1(K)$, it follows that $\gamma = 1$ on $\operatorname{supp}(\varepsilon_{x_1} * \varepsilon_{x_2})$, that is, $\gamma(x) = 1$ for all $x \in \{x_1\} * \{x_2\}$. Thus

$$A(K, \Lambda) * A(K, \Lambda) \subset A(K, \Lambda).$$

Finally we note that $A(K, \Lambda)$ is closed, and this completes the proof of (g).

(h) The proof is similar to that of (f) above. The only part that needs checking is for the analogue of [5], Corollary (23.16), where the fact that $\xi_\Lambda^\vee \in C_0(K)$ is needed; this follows as in the proof of [3], Theorem 2.4.1.

(i) This is proved exactly as for (g) above.

3.2. **Definition.** A family $(\mu_t)_{t>0}$ of measures in $M_+^b(K)$ is called a *semigroup of measures on* K if

(i) $\mu_t(K) \leqq 1$ for all $t > 0$;

(ii) $\mu_{s+t} = \mu_s * \mu_t$ for all $s, t > 0$.

A family $(\rho_\lambda)_{\lambda>0}$ of measures in $M^b_+(K)$ is said to be a *resolvent of measures on K* if

(i) $\lambda\rho_\lambda(K)\leqq 1$ for all $\lambda>0$;

(ii) $\rho_\lambda-\rho_\mu=(\mu-\lambda)\rho_\lambda*\rho_\mu$ for all $\lambda,\mu>0$.

The latter property is known as the *resolvent equation.*

3.3. *Remark.* To avoid trivialities we assume that for any semigroup $(\mu_t)_{t>0}$ or resolvent $(\rho_\lambda)_{\lambda>0}$ of measures, $\mu_t\neq 0$ and $\rho_\lambda\neq 0$ for all $t>0$ and $\lambda>0$ respectively.

We recall that for any $\mu\in M(K)$ the *invariance* or *periodicity set* $L(\mu)$ is given by

$$L(\mu):=\{x\in K:\ \varepsilon_x*\mu=\mu\}.$$

It is known from [3], Corollary 3.3.6 that whenever $\mu\in M^b_+(K)$, $L(\mu)$ is a compact subhypergroup of K. Writing

$$\Lambda:=\{\chi\in K^\wedge:\ \hat{\mu}(\chi)\neq 0\}$$

we easily see that $L(\mu)=A(K,\Lambda)^-$; indeed this follows by applying the Fourier transform to the convolution identity $\varepsilon_x*\mu=\mu$, where $x\in L(\mu)$.

3.4. **Proposition.** *Let $(\mu_t)_{t>0}$ be a semigroup of measures on K. We consider the sets*

$$\Lambda_t:=\{\chi\in K^\wedge:\ \hat{\mu}_t(\chi)\neq 0\}$$

for $t>0$. Then

(a) *Λ_t is independent of $t>0$, thus of the form $\Lambda:=\Lambda_t$.*

(b) *For all $t>0$ we have*

$$L(\mu_t)=A(K,\Lambda)$$

and

$$\Lambda=A(K^\wedge,L(\mu_t)).$$

(c) *Λ is an open (and closed) subhypergroup of $K^\wedge$.*

If $(\mu_t)_{t>0}$ is symmetric then also

(d) $\mathscr{T}_w-\lim_{t\to 0}\mu_t=\omega_{A(K,\Lambda)}$.

Proof. Let $0<t<s$ and choose $n\in\mathbb{N}$ such that $s<nt$. From $\mu_t*\mu_{s-t}=\mu_s$ we deduce $\Lambda_s\subset\Lambda_t$. The reverse inclusion follows similarly. This proves (a).

Now assume that $(\mu_t)_{t>0}$ is symmetric. Applying the Fourier transform to the equality $\mu_t=\mu_t^-$ shows that $\hat{\mu}_t$ is real valued, and the semigroup property ensures that $0\leqq\hat{\mu}_t\leqq 1$ for all $t>0$. This implies

$$\lim_{n\to\infty}\hat{\mu}_{1/n}(\chi)=\lim_{n\to\infty}\hat{\mu}_1(\chi)^{1/n}=\xi_\Lambda(\chi)$$

for all $\chi\in K^\wedge$.

By the remark preceding the proposition,

$$L(\mu_t)=A(K,\Lambda)^-$$

for all $t>0$. Furthermore, $L(\mu_t)$ is a compact subhypergroup of K, whence it follows that $A(K,\Lambda)$ is a compact subhypergroup of K and $L(\mu_t)=A(K,\Lambda)$.

From Lévy's continuity theorem ([3], Theorem 4.6) we now infer the existence of a measure $v \in M^+(K)$ satisfying $\hat{\mu} = \xi_\Lambda$ and

$$\mathscr{T}_w - \lim_{n \to \infty} \mu_{1/n} = v.$$

Similarly one shows that

$$\mathscr{T}_w - \lim_{n \to \infty} \mu_{2/n} = v,$$

from which it follows that $v * v = v$ and, by [3], Theorem 3.3.3, $v = \omega_H$ for some compact subhypergroup H of K. Here we note that by the nontriviality of $(\mu_t)_{t>0}$, $\Lambda \neq \phi$ and $v \neq 0$. We now apply [3], Theorem 3.2.2 in order to obtain

$$\xi_\Lambda = \hat{\omega}_H = \xi_{A(K^\wedge, H)},$$

so that $\Lambda = A(K^\wedge, H)$ and hence, by Proposition 3.1 d, Λ is closed.

We now show that $H = A(K, \Lambda)$. The inclusion $H \subset A(K, \Lambda)$ is clear since for $x \in H$ and $\chi \in \Lambda$, $\chi(x) = 1$ implies $x \in A(K, \Lambda)$. For the reverse inclusion we consider $x \in A(K, \Lambda) = L(\mu_{1/n})$ so that

$$\varepsilon_x * \mu_{1/n} = \mu_{1/n}$$

for all $n \in \mathbb{N}$. It follows that

$$\omega_H = \mathscr{T}_w - \lim_{n \to \infty} \mu_{1/n} = \mathscr{T}_w - \lim_{n \to \infty} \varepsilon_x * \mu_{1/n} = \varepsilon_x * \omega_H$$

and, considering the supports of these measures, $x \in \{x\} * H = H$. This implies $A(K, \Lambda) \subset H$ and we have the desired equality, which takes care of (b) and (d) in the symmetric case. Then (c) follows directly from Proposition 3.1.

Finally, in order to remove the assumption that $(\mu_t)_{t>0}$ be symmetric, we need only observe that

$$\{\chi \in K^\wedge : \hat{\mu}_t(\chi) \neq 0\} = \{\chi \in K^\wedge : (\mu_t * \mu_t^-)^\wedge(\chi) \neq 0\}$$

for each $t > 0$, so that working with the semigroup $(\mu_t * \mu_t^-)_{t>0}$ gives (b) and (c).

In analogy to the preceding proposition we have

3.5. **Proposition.** *Let $(\rho_\lambda)_{\lambda>0}$ be a resolvent of measures on K. We consider the sets*

$$\Lambda_\lambda := \{\chi \in K^\wedge : \hat{\rho}_\lambda(\chi) \neq 0\}$$

for $\lambda > 0$. Then

(a) *Λ_λ is independent of λ, thus of the form $\Lambda := \Lambda_\lambda$.*

(b) *For all $\lambda > 0$ we have*

$$L(\rho_\lambda) = A(K, \Lambda)$$

and

$$\Lambda = A(K^\wedge, L(\rho_\lambda)).$$

(c) *Λ is an open (and closed) subhypergroup of $K^\wedge$.*

(d) $\mathscr{T}_w - \lim_{\lambda \to \infty} \lambda \rho_\lambda = \omega_{A(K,\Lambda)}$.

Proof. The first statement follows immediately from the resolvent equation. Also, for each $\lambda>0$,

$$\hat{\rho}_\lambda(\chi)=\frac{\hat{\rho}_1(\chi)}{1+(\lambda-1)\hat{\rho}_1(\chi)}$$

whenever $\chi\in K^\wedge$, from which it follows that

$$\lim_{\lambda\to\infty}\lambda\hat{\rho}_\lambda=\xi_\Lambda.$$

As in the proof of Proposition 3.4 we obtain $L(\rho_\lambda)=A(K,\Lambda)$, which is a compact subhypergroup of K.

Now putting $\mu=2\lambda$ in the resolvent equation we have

$$\rho_\lambda-\rho_{2\lambda}=\lambda(\rho_\lambda*\rho_{2\lambda})$$

and

$$2\lambda\rho_\lambda-2\lambda\rho_{2\lambda}=\lambda\rho_\lambda*2\lambda\rho_{2\lambda}.$$

By Lévy's continuity theorem ([3], Theorem 4.6) there exists a measure $\nu\in M^+(K)$ such that

$$\lim_{\lambda\to\infty}\lambda\rho_\lambda=\nu.$$

Now, taking the limit as $\lambda\to\infty$ in the above convolution equation, we obtain

$$\nu=2\nu-\nu=\nu*\nu,$$

so that $\nu=\omega_H$ for some compact subhypergroup H of K. Also

$$\xi_\Lambda=\hat{\omega}_H=\xi_{A(K^\wedge,H)}$$

shows that $\Lambda=A(K^\wedge,H)$, and Λ is a (closed) subhypergroup of $K^\wedge$.

It remains to show that $L=A(K,\Lambda)$, but this follows exactly as in the proof of Proposition 3.4.

Now we are in a position to prove the main result of this section, that for every continuous convolution semigroup of measures on K there exists a unique strongly negative definite function on $K^\wedge$.

3.6. **Definition.** A semigroup of measures $(\mu_t)_{t>0}$ on K is called a *continuous convolution semigroup on* K if the mapping $t\to\mu_t$ from $(0,\infty)$ into $M_+^b(K)$ is $\mathcal{T}_w$-continuous.

3.7. **Theorem** (Schoenberg correspondence). *Let $(\mu_t)_{t>0}$ be a continuous convolution semigroup on K and let*

$$\Lambda=\Lambda_t=\{\chi\in K^\wedge:\hat{\mu}_t(\chi)\neq 0\},$$

the subhypergroup of $K^\wedge$ introduced in Proposition 3.4. *Then there exists a uniquely determined function $\psi\in SN(\Lambda)$ such that for all $t>0$,*

$$(\dagger)\qquad \hat{\mu}_t(\chi)=\begin{cases}\exp(-t\psi(\chi)) & \text{if } \chi\in\Lambda,\\ 0 & \text{if } \chi\notin\Lambda.\end{cases}$$

Conversely if $\psi \in SN(\Lambda)$, where Λ is an open subset of $K^\wedge$ of the form $A(K^\wedge, H)$ for some compact subhypergroup H of K, then (†) defines a continuous convolution semigroup $(\mu_t)_{t>0}$ on K with $L(\mu_t)=H$ for all $t>0$.

Proof. At first we consider a continuous convolution semigroup $(\mu_t)_{t>0}$ on K and, for each $\chi \in \Lambda$, the function ϕ_χ: $[0, \infty) \to \mathbb{C}$ defined by

$$\phi_\chi(t) := \begin{cases} \hat{\mu}_t(\chi) & \text{if } t>0, \\ 1 & \text{if } t=0. \end{cases}$$

Then, using the semigroup property of $(\mu_t)_{t>0}$, we deduce that ϕ_χ is continuous. Indeed for $\chi \in \Lambda$,

$$\lim_{t \to 1} |\hat{\mu}_{|t-1|}(\chi) - 1| = \lim_{t \to 1} \frac{1}{|\hat{\mu}_1(\chi)|} |\hat{\mu}_t(\chi) - \hat{\mu}_1(\chi)| = 0,$$

hence $\lim_{t \to 0} \hat{\mu}_t(\chi) = 1$. Since $\hat{\mu}_t(\chi) = 0$ for all $\chi \notin \Lambda$, it is clear that

$$\lim_{t \to 0} \hat{\mu}_t = \xi_\Lambda$$

and it follows as in the proof of Proposition 3.4 that

$$\lim_{t \to 0} \mu_t = \omega_{A(K,\Lambda)}.$$

Also $\phi_\chi(s+t) = \phi_\chi(s) + \phi_\chi(t)$ for all $s, t > 0$, so that there exists $\psi(\chi)$ such that

$$\phi_\chi(t) = \exp(-t\psi(\chi))$$

whenever $\chi \in \Lambda$.

We now show that ψ is a continuous function on $K^\wedge$. Consider the linear functional

$$f \to \rho(f) := \int_0^\infty e^{-t} \mu_t(f)\, dt$$

on $C_c(K)$. Clearly $\rho \geqq 0$,

$$|\rho(f)| \leqq \|f\|_\infty \int_0^\infty e^{-t}\, dt = \|f\|_\infty$$

for all $f \in C_c(K)$, and thus $\rho \in \mathrm{M}_+^b(K)$. Moreover

$$\begin{aligned} \hat{\rho}(\chi) = \rho(\bar{\chi}) &= \int_0^\infty e^{-t} \mu_t(\bar{\chi})\, dt \\ &= \begin{cases} \int_0^\infty \exp[-t(1+\psi(\chi))]\, dt & \text{if } \chi \in \Lambda, \\ 0 & \text{if } \chi \notin \Lambda. \end{cases} \\ &= \begin{cases} \dfrac{1}{1+\psi(\chi)} & \text{if } \chi \in \Lambda, \\ 0 & \text{if } \chi \notin \Lambda. \end{cases} \end{aligned}$$

In particular ψ is continuous on Λ and satisfies $\psi(\mathbb{1}) \geqq 0$ since $\mu_t(K) \leqq 1$ for all $t>0$, and thus $\psi \in SN(\Lambda)$.

To prove the converse implication let $\psi \in SN(\Lambda)$ be given where $\Lambda = A(K^\wedge, H)$ is open and H is a compact subhypergroup of K. By definition there exists, for every $t>0$, a measure $\mu_t \in M^b_+(K)$ satisfying

$$(\dagger\dagger) \qquad \hat{\mu}_t(\chi) = \begin{cases} \exp(-t\psi(\chi)) & \text{if } \chi \in \Lambda, \\ 0 & \text{if } \chi \notin \Lambda. \end{cases}$$

We note that $\hat{\mu}_t(\mathbb{1}) = \exp(-t\psi(\mathbb{1})) \leqq 1$ for all $t>0$ since $\mathbb{1} \in \Lambda$ and, for all $s, t>0$ and $\chi \in \Lambda$,

$$\begin{aligned} \hat{\mu}_s(\chi)\,\hat{\mu}_t(\chi) &= \exp(-s\psi(\chi))\exp(-t\psi(\chi)) \\ &= \exp(-(s+t)\psi(\chi)) \\ &= \hat{\mu}_{s+t}(\chi). \end{aligned}$$

These equalities hold also for all $\chi \notin \Lambda$, so we can apply the injectivity of the Fourier transform ([3], Theorem 3.2.1 (vi)) in order to deduce that $\mu_s * \mu_t = \mu_{s+t}$ whenever $s, t>0$.

We note also that $\Lambda = A(K^\wedge, H)$ is closed. From $(\dagger\dagger)$ it follows that the mapping $t \to \hat{\mu}_t$ from $(0, \infty)$ into $C^b(K^\wedge)$ is continuous with respect to the topology of uniform convergence on compact subsets of $K^\wedge$. This implies that the corresponding mapping $t \to \mu_t$ from $(0, \infty)$ into $M^b_+(K)$ is $\mathscr{T}_w$-continuous. Thus $(\mu_t)_{t>0}$ is a continuous convolution semigroup on K. It is clear that $L(\mu_t) = H$ for each $t>0$.

4. Semigroups and Resolvents

We now show that there exists a bijection between the spaces of continuous convolution semigroups and resolvents of measures on K.

First, given a continuous convolution semigroup $(\mu_t)_{t>0}$ on K, we have by the proof of Theorem 3.7 that $\rho_\lambda(\lambda>0)$ defined by

$$\rho_\lambda(f) := \int_0^\infty e^{-\lambda t}\,\mu_t(f)\,dt$$

for all $f \in C_c(K)$ is a measure in $M^b_+(K)$, with Fourier transform given by

$$\hat{\rho}_\lambda(\chi) = \begin{cases} \dfrac{1}{\lambda + \psi(\chi)} & \text{if } \chi \in \Lambda, \\ 0 & \text{if } \chi \notin \Lambda. \end{cases}$$

Hence, for all $\chi \in \Lambda$,

$$\begin{aligned} \hat{\rho}_\lambda(\chi) - \hat{\rho}_\mu(\chi) &= \frac{1}{\lambda + \psi(\chi)} - \frac{1}{\mu + \psi(\chi)} \\ &= (\mu - \lambda)\,\hat{\rho}_\lambda(\chi)\,\hat{\rho}_\mu(\chi), \end{aligned}$$

and since these equalities clearly hold for all $\chi \notin \Lambda$, the injectivity of the Fourier transform ([3], Theorem 3.2.1 (vi)) yields

$$\rho_\lambda - \rho_\mu = (\mu - \lambda)\rho_\lambda * \rho_\mu,$$

so that $(\rho_\lambda)_{\lambda>0}$ is a resolvent of measures on K.

4.1. **Proposition.** *To every resolvent of measures $(\rho_\lambda)_{\lambda>0}$ on K there corresponds a unique continuous convolution semigroup $(\mu_t)_{t>0}$ on K such that*

$$\rho_\lambda = \int_0^\infty e^{-\lambda t}\mu_t\, dt$$

whenever $\lambda > 0$. Moreover

$$L(\mu_t) = L(\rho_\lambda)$$

holds for all $t, \lambda > 0$.

Proof. From the resolvent equation we deduce, for $\chi \in \Lambda := A(K^\wedge, L(\mu_t))$, $\lambda, \delta > 0$,

$$\psi(\chi) := \frac{1 - \lambda\hat{\rho}_\lambda(\chi)}{\hat{\rho}_\lambda(\chi)} = \frac{1 - \delta\hat{\rho}_\delta(\chi)}{\hat{\rho}_\delta(\chi)},$$

so that the definition of ψ does not depend on the choice of $\lambda > 0$. Clearly ψ is continuous on Λ. Now we have

$$\begin{aligned}\lambda\hat{\rho}_\lambda(\chi)\psi(\chi) &= \lambda(1 - \lambda\hat{\rho}_\lambda(\chi)) \\ &= \lambda(\lambda\hat{\rho}_\lambda(\mathbb{1}) - \lambda\hat{\rho}_\lambda(\chi)) + \lambda(1 - \lambda\hat{\rho}_\lambda(\mathbb{1}))\end{aligned}$$

for all $\chi \in K^\wedge$, $\lambda > 0$. But, as in the proof of Proposition 3.5,

$$\lim_{\lambda\to\infty} \lambda\hat{\rho}_\lambda(\chi) = 1$$

for all $\chi \in \Lambda$. Also, by Proposition 2.4, $\lambda\hat{\rho}_\lambda(\mathbb{1}) - \lambda\hat{\rho}_\lambda \in SN(K^\wedge)$ and, since Λ is an open subhypergroup of $K^\wedge$ with $\xi_\Lambda = \hat{\omega}_{A(K,\Lambda)}$, $\lambda\hat{\rho}_\lambda(\mathbb{1}) - \lambda\hat{\rho}_\lambda \in SN(\Lambda)$. Moreover $1 - \lambda\hat{\rho}_\lambda(\mathbb{1}) \geqq 0$ shows that the constant function $1 - \lambda\hat{\rho}_\lambda(\mathbb{1})$ belongs to $SN(\Lambda)$. Thus ψ is the pointwise limit of functions in $SN(\Lambda)$, and hence itself is an element of $SN(\Lambda)$. Now apply Theorem 3.7 to obtain the result.

4.2. *Résumé.* From Theorem 3.7, Proposition 4.1 and the remarks preceding it, we deduce the useful fact that for every continuous convolution semigroup $(\mu_t)_{t>0}$ on K there exists a uniquely determined strongly negative definite function ψ on Λ and a uniquely determined resolvent of measures $(\rho_\lambda)_{\lambda>0}$ on K. Both objects ψ and $(\rho_\lambda)_{\lambda>0}$ are said to *correspond* to $(\mu_t)_{t>0}$.

4.3. **Proposition.** *Let $(\mu_t)_{t>0}$ be a continuous convolution semigroup on K with corresponding $\psi \in SN(\Lambda)$ and $(\rho_\lambda)_{\lambda>0}$ on K, and assume that $A(K, \Lambda) = \{e\}$. Then for each $\lambda > 0$ the set*

$$S := \operatorname{supp}(\rho_\lambda) = \Big(\bigcup_{t>0} \operatorname{supp}(\mu_t)\Big)^c$$

has the properties:

(i) $e \in S$;

(ii) $S * S \subset S$;

(iii) *The smallest subhypergroup* $[S]$ *containing* S *is* σ*-compact.*

The proof of this proposition will be based on the following lemma.

4.4. **Lemma.** *For every* $\mu \in M_+^b(K)$, $\operatorname{supp}(\mu)$ *is* σ*-compact.*

Proof. For each $n \in \mathbb{N}$ there exists a compact subset C_n of K with $\mu(K \setminus C_n) < \frac{1}{n}$. Then $\bigcup_{n \geqq 1} C_n$ is σ-compact and

$$\operatorname{supp}(\mu) \subset (\bigcup_{n \geqq 1} C_n)^c.$$

But the closure A^c of the σ-compact set $A := \bigcup_{n \geqq 1} C_n$ is σ-compact. Indeed, choose V to be a compact symmetric neighbourhood of e. Then $A \subset A * V$ and hence $A^c \subset A * V$. For if $x \in A^c$ then $(x * V) \cap A \neq \phi$ so that by [7], Lemma 4.1 B, $x \in A * V^- = A * V$. Thus

$$A^c \subset (\bigcup_{n \geqq 1} C_n) * V = \bigcup_{n \geqq 1} C_n * V,$$

and

$$A^c = \bigcup_{n \geqq 1} [(C_n * V) \cap A].$$

But $(C_n * V) \cap A$ is a compact set for each $n \geqq 1$, and this gives the result.

Proof of Proposition 4.3. We have

$$\rho_\lambda(f) = \int_0^\infty e^{-\lambda t} \mu_t(f)\, dt$$

for each $f \in C_c(K)$, and $\mu_t(f) = 0$ for all $t > 0$ if and only if $\rho_\lambda(f) = 0$ $(\lambda > 0)$. Since

$$\lim_{t \to 0} \mu_t(f) = \varepsilon_e(f) = f(e)$$

for all $f \in C_c(K)$ it follows that if $f(e) > 0$ then $\rho_\lambda(f) > 0$, so that $e \in \operatorname{supp}(\rho_\lambda)$.

Let $U \subset K$ be open. Then ρ_λ vanishes on U if and only if μ_t vanishes on U for all $t > 0$. Hence

$$(\bigcup_{t > 0} \operatorname{supp}(\mu_t))^c = \operatorname{supp}(\rho_\lambda)$$

which, by Lemma 4.4, is a σ-compact set. We now apply [7], Lemma 3.2 F, to obtain

$$\operatorname{supp}(\mu_{s+t}) = \operatorname{supp}(\mu_s * \mu_t) = (\operatorname{supp}(\mu_s) * \operatorname{supp}(\mu_t))^c \qquad (s, t > 0).$$

Writing $N := \bigcup_{t > 0} \operatorname{supp}(\mu_t)$ we have

$$\operatorname{supp}(\mu_s) * \operatorname{supp}(\mu_t) \subset N \qquad (s, t > 0)$$

which shows that $N * N \subset N$ and

$$N^c * N^c \subset (N * N)^c \subset N^c.$$

An application of [7], Lemma 10.1 C yields the remaining assertion.

5. Periodicity of Strongly Positive Definite Functions

The idea of *invariance* or *periodicity set* can be extended to $f \in C(K^\wedge)$ as follows:

$$L(f) := \{\chi \in K^\wedge : \varepsilon_\chi * f = f\}.$$

5.1. **Proposition.** *Assume that $K^\wedge$ is a hypergroup under pointwise multiplication and let $\mu \in M^b(K)$. Then*

(a) $L(\hat{\mu}) = A(K^\wedge, \operatorname{supp}(\mu))$.

If $\phi \in SP(K^\wedge)$ with $\phi = \hat{\sigma}$ for some $\sigma \in M^b_+(K)$, then

(b) $L(\phi) = A(K^\wedge, \operatorname{supp}(\sigma)) = \{\chi \in K^\wedge : \phi(\chi) = \phi(\mathbb{1})\}$.

If $\psi \in SN(K^\wedge)$ then

(c) *for all $t > 0$,*

$$\begin{aligned} L(e^{-t\psi}) &= \{\chi \in K^\wedge : e^{-t\psi(\chi)} = e^{-t\psi(\mathbb{1})}\} \\ &= \{\chi \in K^\wedge : \psi(\chi) = \psi(\mathbb{1})\}. \end{aligned}$$

Proof. (a) Here we apply the assumption that $K^\wedge$ is a hypergroup. Using [7], Lemma 12.1 G, Lemma 12.1 E and Lemma 4.2 H we obtain

$$\begin{aligned} (\gamma\mu)^\wedge(\chi) &= \int_K \bar{\chi}\gamma \, d\mu = \int_{K^\wedge} \hat{\mu}^- \, d(\varepsilon_{\chi^-} * \varepsilon_\gamma) \\ &= \int_{K^\wedge} \varepsilon_{\gamma^-} * \hat{\mu}^- \, d\varepsilon_{\chi^-} = \varepsilon_\gamma * \hat{\mu}(\chi) \end{aligned}$$

for all $\chi, \gamma \in K^\wedge$, whence

$$\begin{aligned} \gamma \in L(\hat{\mu}) &\Leftrightarrow \gamma\mu = \mu \\ &\Leftrightarrow \gamma = 1 \text{ on } \operatorname{supp}(\mu) \\ &\Leftrightarrow \gamma \in A(K^\wedge, \operatorname{supp}(\mu)). \end{aligned}$$

(b) The first equality is the content of (a). As for the second, consider $\gamma \in L(\phi)$. Applying Proposition 3.1(f), we see that $\gamma^- \in L(\phi)$ and $\varepsilon_{\gamma^-} * \phi = \phi$. Hence

$$\begin{aligned} \phi(\mathbb{1}) = \varepsilon_{\gamma^-} * \phi(\mathbb{1}) &= \int_{K^\wedge} \phi(\chi^- * \mathbb{1}) \, d\varepsilon_{\gamma^-}(\chi) \\ &= \int_{K^\wedge} \left(\int_{K^\wedge} \phi \, d\varepsilon_{\chi^-} * \varepsilon_{\mathbb{1}} \right) d\varepsilon_{\gamma^-}(\chi) \\ &= \int_{K^\wedge} \phi(\chi^-) \, d\varepsilon_{\gamma^-}(\chi) \\ &= \phi(\gamma). \end{aligned}$$

To complete the proof, suppose that $\phi(\gamma)=\phi(\mathbb{1})$ so that $\hat{\sigma}(\gamma)=\hat{\sigma}(\mathbb{1})$. We can assume that $\phi\neq 0$ and, by normalising if necessary, that $\sigma\in M^1(K)$. Thus

$$\int_K \bar{\gamma}\, d\sigma = 1.$$

Since $\|\gamma\|_\infty = 1$ it follows that $\bar{\gamma}=1$ on $\mathrm{supp}(\sigma)$, which is equivalent to saying that $\gamma\in A(K^\wedge, \mathrm{supp}(\sigma))$.

(c) The first equality is just (b); the second one is clear.

5.2. *Remark.* Examining the proof of Proposition 5.1(b) we see that for each $g\in C(K^\wedge)$ we have

$$L(g)\subset\{\chi\in K^\wedge : g(\chi^-)=g(\mathbb{1})\}.$$

5.3. **Theorem.** *For every* $\psi\in SN(K^\wedge)$,

$$\bigcap_{t>0} L(e^{-t\psi})\subset L(\psi).$$

Proof. Suppose that $\varepsilon_\gamma * e^{-t\psi}=e^{-t\psi}$ for all $t>0$. Then

$$\varepsilon_\gamma * (e^{-t\psi}-1)=\varepsilon_\gamma * e^{-t\psi}-\varepsilon_\gamma * 1 = e^{-t\psi}-1,$$

since clearly for all $\chi\in K^\wedge$,

$$\begin{aligned}\varepsilon_\gamma * 1(\chi) &= \int_{K^\wedge} 1(\eta^- * \chi)\, d\varepsilon_\gamma(\eta)\\ &= \int_{K^\wedge}\Big(\int_{K^\wedge} 1\, d(\varepsilon_{\eta^-} * \varepsilon_\chi)\Big) d\varepsilon_\gamma(\eta)\\ &= \int_{K^\wedge} d\varepsilon_\gamma(\eta) = 1.\end{aligned}$$

Furthermore, putting $g_t := \frac{1}{t}(e^{-t\psi}-1)$ for all $t>0$, we obtain for all $\chi\in K^\wedge$,

$$\begin{aligned}-\psi(\chi) &= \lim_{t\downarrow 0}\left(\varepsilon_\gamma * \frac{e^{-t\psi}-1}{t}\right)(\chi)\\ &= \lim_{t\downarrow 0}\int_{K^\wedge} g_t(\eta^- * \chi)\, d\varepsilon_\gamma(\eta)\\ &= \lim_{t\downarrow 0}\int_{K^\wedge}\Big(\int_{K^\wedge} g_t\, d(\varepsilon_{\eta^-} * \varepsilon_\chi)\Big) d\varepsilon_\gamma(\eta).\end{aligned}$$

Now we shall apply Lebesgue's dominated convergence theorem twice, to the outer and to the inner integrals of the last equality.

Since the function

$$\eta \to f_t(\eta) := \int_{K^\wedge} g_t\, d(\varepsilon_{\eta^-} * \varepsilon_\chi)$$

satisfies

$$\begin{aligned}|f_t(\eta)| &\leqq \int_{K^\wedge} |\psi|\, e^{t|\psi|}\, d(\varepsilon_{\eta^-} * \varepsilon_\chi)\\ &\leqq 2\max\{|\psi|(\gamma) : \gamma\in\{\eta^-\} * \{\chi\}\}\end{aligned}$$

for sufficiently small t (with $t \max\{|\psi|(\gamma): \gamma \in \{\eta^-\} * \{\chi\}\} \leqq \log 2$ say), and $2 \max\{|\psi|(\gamma): \gamma \in \{\eta^-\} * \{\chi\}\}$ is integrable with respect to ε_γ, we obtain

$$-\psi(\chi) = \int_{K^\wedge} (\lim_{t \downarrow 0} \int_{K^\wedge} g_t \, d(\varepsilon_{\eta^-} * \varepsilon_\chi)) \, d\varepsilon_\gamma(\eta).$$

Moreover we have

$$|g_t| \leqq |\psi| \, e^{t|\psi|} \leqq 2|\psi|$$

on $\{\eta^-\} * \{\chi\}$ for all $t > 0$ sufficiently small, and ψ is integrable with respect to $\varepsilon_{\eta^-} * \varepsilon_\chi$, since ψ is continuous on the compact set $\{\eta^-\} * \{\chi\} = \operatorname{supp}(\varepsilon_{\eta^-} * \varepsilon_\chi)$. Therefore

$$\begin{aligned} -\psi(\chi) &= \int_{K^\wedge} \Big(\int_{K^\wedge} (\lim_{t \downarrow 0} g_t) \, d(\varepsilon_{\eta^-} * \varepsilon_\chi) \Big) \, d\varepsilon_\gamma(\eta) \\ &= \int_{K^\wedge} \Big(\int_{K^\wedge} (-\psi) \, d(\varepsilon_{\eta^-} * \varepsilon_\chi) \Big) \, d\varepsilon_\gamma(\eta) \\ &= -\varepsilon_\gamma * \psi(\chi) \end{aligned}$$

for all $\chi \in K^\wedge$, whence $\gamma \in L(\psi)$.

The following result generalises part of [2], Proposition 8.27 to hypergroups.

5.4. **Theorem.** *Let $\psi \in SN(K^\wedge)$ with corresponding continuous convolution semigroup $(\mu_t)_{t>0}$ on K. Then*

(i) *for all $t > 0$,*

$$L(e^{-t\psi}) = L(\psi) = \{\chi \in K^\wedge : \psi(\chi^-) = \psi(\mathbb{1})\},$$

and

(ii) $L(\psi) = A(K^\wedge, H)$ *as well as*

$$H \subset A(K, L(\psi)),$$

where $H := [\bigcup_{t>0} \operatorname{supp}(\mu_t)]$.

Proof. By Proposition 5.1(c) we obtain

$$\{\chi \in K^\wedge : \psi(\chi^-) = \psi(\mathbb{1})\} = \bigcap_{t>0} L(e^{-t\psi})$$

and, by Theorem 5.3 and Remark 5.2, we have the inclusions

$$\begin{aligned} \bigcap_{t>0} L(e^{-t\psi}) &\subset L(\psi) \\ &\subset \{\chi \in K^\wedge : \psi(\chi^-) = \psi(\mathbb{1})\} \end{aligned}$$

respectively. This proves (i).

Concerning the proof of (ii) we note that for all $t > 0$,

$$\hat{\mu}_t = e^{-t\psi}.$$

Proposition 5.1(b) implies that

$$L(\psi) = L(e^{-t\psi}) = A(K^\wedge, \operatorname{supp}(\mu_t))$$

for all $t>0$, whence

$$\begin{aligned} L(\psi) &= \bigcap_{t>0} A(K^\wedge, \operatorname{supp}(\mu_t)) \\ &= A(K^\wedge, \bigcup_{t>0} \operatorname{supp}(\mu_t)) \\ &= A(K^\wedge, H) \end{aligned}$$

where, for the last equality, Proposition 3.1(a) has been used.

For the final statement we conclude from $L(\psi)=A(K^\wedge, H)$ that for $\chi\in L(\psi)$, $\chi(x)=1$ for all $x\in H$. This just says that $H\subset A(K, L(\psi))$, which is the desired inclusion.

6. Positive Definite Measures

In this section we consider positive definite measures on hypergroups. We follow the expositions of [1] and [2] where the approach starts from Bochner's theorem which is available for arbitrary commutative hypergroups: in terms of (bounded) positive definite functions on K in [7], Theorem 12.3 B, and in terms of strongly positive definite functions on $K^\wedge$ in [9], Theorem 1.

By $P(K)$ we denote as usual the cone of all continuous positive definite functions on K in the sense of [3], Sect. 3.1.

6.1. **Proposition.** *A function $\phi\in C(K)$ is positive definite if and only if for each $f\in C_c(K)$,*

$$\int_K \phi(x)\, f*f^\sim(x)\, d\omega_K(x)\geqq 0.$$

The proof is carried out as in [2], *Proposition* 4.1.

6.2. **Definition.** A measure $\mu\in M(K)$ is said to be *positive definite* if, for all $f\in C_c(K)$,

$$\langle \mu, f*f^\sim\rangle := \int_K f*f^\sim(x)\, d\mu(x)\geqq 0.$$

The set $M_p(K)$ of all positive definite measures on K is a $\mathcal{T}_v$-closed convex cone in $M(K)$ that is stable with respect to involution and conjugation. Clearly $\phi\in C(K)$ is the density of a positive definite measure on K if and only if $\phi\in P(K)$.

We recall the notions of *Fourier transform* and *Fourier cotransform* of a measure $\mu\in M^b(K)$, given by

$$(\mathcal{F}_K\mu)(\chi) := \hat{\mu}(\chi) := \int_K \overline{\chi(x)}\, d\mu(x)$$

and

$$(\bar{\mathcal{F}}_K\mu)(\chi) := \hat{\mu}^-(\chi) := \int_K \chi(x)\, d\mu(x)$$

respectively, whenever $\chi\in K^\wedge$, with corresponding definitions for measures defined on $K^\wedge$. Note that $\bar{\mathcal{F}}_{K^\wedge}\mu=\check{\mu}$, the inverse Fourier transform of μ. We have the following analogue of Bochner's theorem.

6.3. **Theorem.** *The Fourier cotransform $\bar{\mathscr{F}}_{K^\wedge}$ is a bijection between the cones $M_+^b(K^\wedge)$ and $P(K)$.*

Proof. This is given in [7], Sect. 12.3.

Let $\phi \in P(K)$. By Bochner's theorem there exists exactly one measure $\sigma \in M_+^b(K^\wedge)$ such that

$$(\bar{\mathscr{F}}_{K^\wedge}\sigma)(x) = \int_{K^\wedge} \chi(x)\, d\sigma(x) = \phi(x)$$

for all $x \in K$. Clearly $\phi(e) = \sigma(K^\wedge)$.

6.4. **Definition.** By the inversion formula ([7], Theorem 12.2 C) it is natural to call σ the *Fourier transform* of ϕ; we write

$$\mathscr{F}_K \phi := \hat{\phi} := \sigma.$$

6.5. **Theorem** (P. Lévy). *$\mathscr{F}_K$ is a homeomorphism between the cones $P(K)$ and $M_+^b(K^\wedge)$, where $P(K)$ carries the topology of compact convergence and $M_+^b(K^\wedge)$ the weak topology $\mathscr{T}_w$.*

Proof. We restrict ourselves to showing that $\mathscr{F}_K$ is continuous; for the rest of the proof see [2], Theorem 3.13 or our studies in [3], Sect. 4.

Let (ϕ_α) be a filtered family of functions in $P(K)$ such that

$$\lim_\alpha \phi_\alpha = \phi \in P(K).$$

Let $\mu_\alpha := \mathscr{F}_K \phi_\alpha$ for all α and $\mu := \mathscr{F}_K \phi$ be the measures in $M_+^b(K^\wedge)$ corresponding (via Bochner's theorem) to ϕ_α and ϕ respectively. Since

$$\lim_\alpha \mu_\alpha(K^\wedge) = \lim_\alpha \phi_\alpha(e) = \phi(e) = \mu(K^\wedge),$$

it suffices to show that $\mathscr{T}_v - \lim_\alpha \mu_\alpha = \mu$.

Let $h \in C_c(K^\wedge)$ and $\varepsilon > 0$. By the Riemann-Lebesgue lemma ([7], Theorem 7.3 H) there exists a function $f \in C_c(K)$ such that

$$\|h - \mathscr{F}_K f\|_\infty < \varepsilon.$$

We obtain

$$\begin{aligned}
\left|\int_{K^\wedge} h\, d\mu_\alpha - \int_{K^\wedge} h\, d\mu\right| &\leqq \left|\int_{K^\wedge} (h - \mathscr{F}_K f)\, d\mu_\alpha\right| \\
&\quad + \left|\int_{K^\wedge} \mathscr{F}_K f\, d\mu_\alpha - \int_{K^\wedge} \mathscr{F}_K\, d\mu\right| + \left|\int_{K^\wedge} (\mathscr{F}_K f - h)\, d\mu\right| \\
&\leqq \varepsilon(\mu_\alpha(K^\wedge) + \mu(K^\wedge)) \\
&\quad + \int_K |\phi_\alpha(x) - \phi(x)|\, f(x^-)\, d\omega_K(x),
\end{aligned}$$

where the equalities

$$\int_{K^\wedge} \mathscr{F}_K f\, d\mu_\alpha = \int_{K^\wedge} (\mathscr{F}_K f)\, d(\mathscr{F}_K \phi_\alpha) = \int_K \phi_\alpha f^-\, d\omega_K$$

([7], Lemma 12.1 G) have been used, and thus

$$\lim_{\alpha} \int_{K^\wedge} h\, d\mu_\alpha = \int_{K^\wedge} h\, d\mu,$$

as desired.

6.6. **Theorem.** *To every measure $\mu \in M_p(K)$ corresponds a unique measure $\sigma \in M^+(K^\wedge)$ such that for all $f \in C_c(K)$ and for all $x \in K$*

(i) $\int_K |\mathscr{F}_K f|^2\, d\sigma < \infty$

and

(ii) $\mu * f * f^\sim(x) = \int_{K^\wedge} \chi(x)\, |(\mathscr{F}_K f)(\chi)|^2\, d\sigma(\chi)$

6.7. **Definition.** The measure σ of Theorem 6.6 is called the *Fourier transform* of μ and will be denoted by

$$\sigma := \mathscr{F}_K \mu := \hat{\mu}.$$

Proof of Theorem 6.6. First we note that for every $f \in C_c(K)$ the convolution $\mu * f * f^\sim$ belongs to $P(K)$. Indeed for all $g \in C_c(K)$,

$$\int \mu * f * f^\sim(x)\, g * g^\sim(x)\, d\omega_K(x) = \langle \mu, (f^- * g) * (f^- * g)^\sim \rangle \geqq 0$$

and, by Proposition 6.1, this gives the assertion.

Now we infer from Theorem 6.3 that there exists a measure $\sigma_f \in M_+^b(K^\wedge)$ satisfying

$$\mu * f * f^\sim = \bar{\mathscr{F}}_{K^\wedge} \sigma_f.$$

Since

$$\bar{\mathscr{F}}_{K^\wedge}(|\mathscr{F}_K f|^2\, \sigma_g) = \bar{\mathscr{F}}_{K^\wedge}(|\mathscr{F}_K g|^2\, \sigma_f) = \mu * f * f^\sim * g * g^\sim,$$

we obtain

$$(*) \qquad |\mathscr{F}_K f|^2\, \sigma_g = |\mathscr{F}_K g|^2\, \sigma_f.$$

But for every $\sigma \in M^+(K^\wedge)$ satisfying (i) and (ii) we have

$$|\mathscr{F}_K f|^2\, \sigma = \sigma_f$$

whenever $f \in C_c(K)$. From this we see that $\int_{K^\wedge} \phi\, d\sigma$ is uniquely determined for every $\phi \in C_c(K^\wedge)$. We just have to choose a function $f \in C_c(K)$ such that $\hat{f}(\chi) \neq 0$ for all $\chi \in \mathrm{supp}(\phi)$ (see [7], Theorem 7.3H), and we obtain

$$\int_{K^\wedge} \phi\, d\sigma = \int_{K^\wedge} \frac{\phi}{|\hat{f}|^2}\, d\sigma_f,$$

where $\dfrac{\phi}{|\hat{f}|^2}$ denotes the function in $C_c(K^\wedge)$ given by

$$\chi \to \begin{cases} \dfrac{\phi(\chi)}{|\hat{f}(\chi)|^2} & \text{if } \hat{f}(\chi) \neq 0, \\ 0 & \text{otherwise.} \end{cases}$$

We now define $\int_{K^\wedge} \phi \, d\sigma$ as in the preceding paragraph which, by (*), is independent of f. Moreover $\phi \to \sigma(\phi) = \int_{K^\wedge} \phi \, d\sigma$ is a positive linear functional on $C_c(K^\wedge)$, thus a measure $\sigma \in M^+(K^\wedge)$. It remains to show that σ satisfies (i) and (ii) of the theorem. For this it suffices to establish the equality

$$\sigma_f = |\mathscr{F}_K f|^2 \sigma$$

valid for all $f \in C_c(K)$. Indeed let $\phi \in C_c(K^\wedge)$ and $g \in C_c(K)$ such that $\hat{g}(\chi) \neq 0$ for all $\chi \in \mathrm{supp}(\phi)$. Then (*) together with the definition of $\sigma(\phi)$ imply

$$\begin{aligned} \sigma(|\mathscr{F}_K f|^2 \phi) &= \int_{K^\wedge} \frac{|\mathscr{F}_K f|^2 \phi}{|\mathscr{F}_K g|^2} d\sigma_g \\ &= \int_{K^\wedge} \frac{\phi}{|\mathscr{F}_K g|^2} |\mathscr{F}_K g|^2 d\sigma_f \\ &= \int_{K^\wedge} \phi \, d\sigma_f, \end{aligned}$$

and this is the desired assertion.

6.8. **Theorem.** *Let $\mu \in M_p(K)$ and $\sigma \in M^+(K^\wedge)$. The following statements are equivalent:*

(i) $\sigma = \mathscr{F}_K \mu$.

(ii) *For all $f \in C_c(K)$ we have*

$$\int_K f * f^\sim \, d\mu = \int_{K^\wedge} |\bar{\mathscr{F}}_K f|^2 \, d\sigma = \int_{K^\wedge} \bar{\mathscr{F}}_K f \, \overline{\bar{\mathscr{F}}_K f} \, d\sigma.$$

Proof. (i)$\Rightarrow$(ii) follows from (ii) of Theorem 6.6 if we choose f to be f^- and $x = e$.

(ii)$\Rightarrow$(i). Polarization of the identity in (ii) yields

$$\langle \mu, f * g^\sim \rangle = \int (\bar{\mathscr{F}}_K f) \, \overline{\bar{\mathscr{F}}_K g} \, d\sigma$$

for all $f, g \in C_c(K^\wedge)$. Replacing f by $\varepsilon_x * \bar{f}$ and g by $\bar{f}$ we obtain the assertion.

6.9. **Corollary.** *The Fourier transform $\mathscr{F}_K$: $M_p(K) \to M^+(K^\wedge)$ is injective.*

The proof of Corollary 6.9 follows from the theorem by polarization.

6.10. *Remark.* Since $\varepsilon_e \in M_p(K)$, Theorem 6.6 implies the existence of $\sigma \in M^+(K^\wedge)$ such that

$$f * f^\sim(x) = \int_{K^\wedge} \chi(x) \, |\hat{f}(\chi)|^2 \, d\sigma(\chi)$$

for all $f \in C_c(K)$ and $x \in K$; in particular taking $x = e$,

$$\int_K |f(x)|^2 \, d\omega_K(x) = \int_{K^\wedge} |\hat{f}(\chi)|^2 \, d\sigma(\chi)$$

for all $f \in C_c(K)$ and, since $K^\wedge$ is assumed to be a hypergroup under pointwise multiplication, [7], Theorem 12.4A implies that $\sigma = \omega_{K^\wedge}$.

6.11. **Proposition.** *Let $\mu\in M^b(K)$. Then $\mu\in M_p(K)$ if and only if $\hat{\mu}(\chi)\geqq 0$ for all $\chi\in K^\wedge$, and in this case*

$$\mathscr{F}_K\mu=\hat{\mu}\,\omega_{K^\wedge}.$$

Proof. Since $\mu\in M^b(K)$ we have $|\hat{\mu}(\chi)|\leqq\|\mu\|$ for all $\chi\in K^\wedge$ and therefore

$$\int_{K^\wedge}|\hat{f}(\chi)|^2\,|\hat{\mu}(\chi)|\,d\omega_{K^\wedge}(\chi)\leqq\|\mu\|\int|\hat{f}(\chi)|^2\,d\omega_{K^\wedge}(\chi)<\infty$$

whenever $f\in C_c(K)$. Remark 6.10 with σ replaced by $\omega_{K^\wedge}$ implies that

$$\mu*f*f^{\sim}(x)=\int_{K^\wedge}\chi(x)\,|\hat{f}(\chi)|^2\,\hat{\mu}(\chi)\,d\omega_{K^\wedge}(\chi)$$

holds for all $f\in C_c(K)$ and all $x\in K$. This shows that the measure $\hat{\mu}\,\omega_{K^\wedge}$ satisfies the conditions (i) and (ii) of Theorem 6.6; thus $\hat{\mu}\,\omega_{K^\wedge}\in M^+(K^\wedge)$ and consequently, $\hat{\mu}\geqq 0$ on $K^\wedge$.

If conversely $\hat{\mu}\geqq 0$ on $K^\wedge$ the above formula shows that

$$\langle\mu, f*f^{\sim}\rangle=\int|\bar{\mathscr{F}}_K f(x)|^2\,\hat{\mu}(\chi)\,d\omega_{K^\wedge}(\chi)\geqq 0$$

for all $f\in C_c(K)$, and hence that $\mu\in M_p(K)$.

6.12. *Remark.* All of the results of this section continue to hold with the roles of K, $K^\wedge$ interchanged, provided that we know that every bounded continuous positive definite function is strongly positive definite. This condition is equivalent to K being a strong hypergroup; see [9], Proposition 4.

7. Transience of Convolution Semigroups

Let $(\mu_t)_{t>0}$ be a continuous convolution semigroup on K with $\mu_0:=\mathscr{T}_w-\lim_{t\to 0}\mu_t=\varepsilon_e$. By the Résumé 4.2 there correspond to $(\mu_t)_{t>0}$ a strongly negative definite function ψ on $K^\wedge$ and a resolvent of measures $(\rho_\lambda)_{\lambda>0}$ on K. For every $f\in C_c^+(K)$ we have

$$\lim_{\lambda\to 0}\rho_\lambda(f)=\int_0^\infty\mu_t(f)\,dt\leqq\infty.$$

If $\lim_{\lambda\to 0}\rho_\lambda(f)<\infty$ for all $f\in C_c^+(K)$ then the relationship

$$\kappa(f):=\lim_{\lambda\to 0}\rho_\lambda(f)=\int_0^\infty\mu_t(f)\,dt$$

valid for all $f\in C_c^+(K)$ defines the *potential measure* κ of $(\mu_t)_{t>0}$. Clearly $\kappa\in M^+(K)$.

7.1. **Definition.** $(\mu_t)_{t>0}$ is called *transient* if

$$\lim_{\lambda\to 0}\rho_\lambda(f)<\infty$$

for all $f\in C_c^+(K)$. Otherwise $(\mu_t)_{t>0}$ is said to be *recurrent*.

7.2. **Proposition.** *For any transient convolution semigroup* $(\mu_t)_{t>0}$ *(with* $\mu_0 := \varepsilon_e$*) on* K *we have*

$$\operatorname{supp}(\kappa) = \operatorname{supp}(\rho_\lambda) = (\bigcup_{t>0} \operatorname{supp}(\mu_t))^c$$

whenever $\lambda > 0$, *and* $S := \operatorname{supp}(\kappa)$ *has the properties* (i) *to* (iii) *of Proposition* 4.3.

Proof. This is an immediate consequence of Proposition 4.3 (and its proof), since $\mu_0 = \varepsilon_e$.

7.3. *Remark.* Let $(\mu_t)_{t>0}$ be a continuous convolution semigroup on K with $\mu_0 := \varepsilon_e$. Assume that there exists a compact subset L of K such that $\operatorname{supp}(\mu_t) \subset L$ for all $t > 0$. Then $(\mu_t)_{t>0}$ is transient if and only if for the strongly negative definite function ψ corresponding to $(\mu_t)_{t>0}$, $\psi(\mathbb{1}) > 0$.

7.4. *Discussion.* (Probabilistic interpretation of continuous convolution semigroups.)

Markov processes $\mathfrak{X}$ with a hypergroup K as state space are given by their semigroups $(N_t)_{t>0}$ of transition functions $(x, B) \to N_t(x, B)$ on $K \times \mathcal{L}(K)$, where $\mathcal{L}(K)$ denotes the Borel σ-algebra of K. Here $N_t(x, B)$ denotes the probability that $\mathfrak{X}$ starting in x at time t will enter the set B at time t.

$\mathfrak{X}$ is called *translation invariant* if for all $(x, B) \in K \times \mathcal{L}(K)$ and for all $y \in K$ and $t \geqq 0$,

$$N_t(x * y, B * y) = N_t(x, B).$$

It is well-known that there is a one-to-one correspondence between the set of all translation invariant Markov processes and the set of all continuous convolution semigroups on K, given by

$$N_t(x, B) = \varepsilon_x * \mu_t(B) = \mu_t(B * x^-)$$

for all $(x, B) \in K \times \mathcal{L}(K)$, $t \geqq 0$.

In the special case of a *random walk* $\mathfrak{X}$ on K it suffices to consider one single translation invariant transition kernel N on K in order to determine $\mathfrak{X}$. In this situation we have a one-to-one correspondence between the set of all translation invariant transition kernels and the set of all probability measures on K, given by

$$N(x, B) = \varepsilon_x * \mu(B) = \mu(B * x^-)$$

for all $(x, B) \in K \times \mathcal{L}(K)$.

For a more detailed description of random walks on a hypergroup, in particular of the translation operator involved, see [6], Chap. IV.

7.5. *Example.* For any measure $\mu \in M_+^b(K)$ with $\mu(K) \leqq 1$ we introduce the *Poisson semigroup* $(\mu_t)_{t>0}$ on K given by

$$\mu_t := e^{-t} \exp(t\,\mu)$$

for all $t > 0$. (Compare this definition with that of [6], Example 9.3.) An easy computation ([2], Lemma 13.5) shows that $(\mu_t)_{t>0}$ is transient if and only if $\sum_{n \geqq 1} \mu^n < \infty$ (in the sense of $\mathcal{T}_v$). In this case we have for the potential measure κ

of $(\mu_t)_{t>0}$ that

$$\kappa = \sum_{n \geqq 0} \mu^n.$$

7.7. *Notation.* Let $\psi \in SN(K^\wedge)$. We say that $\frac{1}{\psi}$ is *integrable on a neighbourhood* Ω *of* $\mathbb{1}$ if $\psi \neq 0$ $\omega_{K^\wedge}$-a.e. on Ω and $\frac{1}{\psi}$ (defined $\omega_{K^\wedge}$-a.e. on Ω) is $\omega_{K^\wedge}$-integrable on Ω. A similar definition will apply to local integrability.

7.8. **Proposition** (Generalization of Proposition 2.5). *Let* $\psi \in SN(K^\wedge)$ *and suppose that* $\frac{1}{\psi}$ *is integrable on an open, relatively compact neighbourhood* Ω *of* $\mathbb{1}$. *Then*

(a) $\frac{1}{\psi}$ *is* $\omega_{K^\wedge}$*-locally integrable*

and

(b) $\frac{1}{\psi}\omega_{K^\wedge} \in M_p(K^\wedge)$.

Proof. We only show (a); the proof of (b) is carried out exactly as in the proof of [2], Proposition 13.13.

Let Ξ be a compact subset of $K^\wedge$. If $\psi(\chi) \neq 0$ for all $\chi \in \Xi$ then

$$\int_\Xi \frac{1}{|\psi(\chi)|} d\omega_{K^\wedge}(\chi) < \infty.$$

Now suppose $\psi(\chi) = 0$ for some $\chi \in \Xi$. Clearly

$$\Xi_0 := \{\chi \in \Xi : \psi(\chi) = 0\}$$

is a compact subset of $K^\wedge$, and thus there exist characters $\chi_1, \ldots, \chi_n \in \Xi_0$ satisfying $\Xi_0 \subset \Omega' := \bigcup_{i=1}^n \chi_i * \Omega$, where Ω denotes an open, relatively compact neighbourhood of $\mathbb{1}$ with the property that $\frac{1}{\psi}$ is integrable on Ω. We note that $\Lambda := \Xi \setminus \Omega'$ is a compact subset of Ξ with $\psi(\chi) \neq 0$ for all $\chi \in \Lambda$. This implies that $\psi \neq 0$ $\omega_{K^\wedge}$-a.e. on Ξ, and

$$\begin{aligned}\int_\Xi \frac{1}{|\psi(\chi)|} d\omega_{K^\wedge}(\chi) &\leqq \int_\Lambda \frac{1}{|\psi(\chi)|} d\omega_{K^\wedge}(\chi) + \int_{\Omega'} \frac{1}{|\psi(\chi)|} d\omega_{K^\wedge}(\chi) \\ &\leqq \int_\Lambda \frac{1}{|\psi(\chi)|} d\omega_{K^\wedge}(\chi) + n \int_\Omega \frac{1}{|\psi(\chi)|} d\omega_{K^\wedge}(\chi),\end{aligned}$$

where the second inequality follows from the fact that $\Xi_0 \subset L(\psi)$, which has been established in Theorem 5.4(i). But now the assumption can be used in order to finish the proof.

7.9. **Theorem** (Necessary condition for transience). *Let* $(\mu_t)_{t>0}$ *be a continuous convolution semigroup on* K *such that* $\mu_0 = \varepsilon_e$, *with corresponding strongly nega-*

tive definite function ψ on $K^\wedge$. If $(\mu_t)_{t>0}$ is transient then $\operatorname{Re}\frac{1}{\psi}$ *is locally $\omega_{K^\wedge}$-integrable.*

Proof. Without loss of generality we can assume that $\psi(\mathbb{1})=0$.

First we show that $\psi \neq 0$ locally $\omega_{K^\wedge}$-a.e. on $K^\wedge$. Indeed by Theorem 5.4 we know that the set

$$L(\psi) := \{\chi \in K^\wedge : \psi(\chi^-)=0\}$$

coincides with the subhypergroup $A(K^\wedge, H)$ of $K^\wedge$, where $H := [\bigcup_{t>0} \operatorname{supp}(\mu_t)]$. Moreover, by the same theorem, we have

$$H \subset A(K, L(\psi)).$$

Since by assumption $(\mu_t)_{t>0}$ is transient, $A(K, L(\psi))$ is not compact (see Remark 7.3). We now infer from Proposition 3.1(h) that $L(\psi)$ is a local $\omega_{K^\wedge}$-null set. But this is exactly the assertion we want.

For the main part of the proof we choose a compact subset Ξ of $K^\wedge$ and a function $f \in C_c^+(K)$ satisfying $|\bar{\mathscr{F}}_K f| \geqq 1$ on Ξ (whose existence follows from [7], Theorem 7.3H). An application of Fatou's theorem, Theorem 6.8 and the transience of $(\mu_t)_{t>0}$ yields

$$\begin{aligned}
\int_\Xi \operatorname{Re}\frac{1}{\psi}\, d\omega_{K^\wedge} &= \int_\Xi \lim_{\lambda\to 0} \operatorname{Re}\frac{1}{\lambda+\psi}\, d\omega_{K^\wedge} \\
&\leqq \int_{K^\wedge} \varliminf_{\lambda\to 0} |\bar{\mathscr{F}}_K f|^2 \operatorname{Re}\frac{1}{\lambda+\psi}\, d\omega_{K^\wedge} \\
&\leqq \varliminf_{\lambda\to 0} \int_{K^\wedge} |\bar{\mathscr{F}}_K f|^2 \operatorname{Re}\frac{1}{\lambda+\psi}\, d\omega_{K^\wedge} \\
&= \varliminf_{\lambda\to 0} \tfrac{1}{2}\int_K f * f^\sim\, d(\rho_\lambda + \rho_\lambda^-) \\
&= \tfrac{1}{2}\int_K f * f^\sim\, d(\kappa+\kappa^-) < \infty,
\end{aligned}$$

where $(\rho_\lambda)_{\lambda>0}$ and κ denote the resolvent and the potential measure corresponding to $(\mu_t)_{t>0}$ respectively. One just notes that $\frac{1}{2}(\rho_\lambda+\rho_\lambda^-)$ is a measure in $M_p(K)$ with Fourier transform

$$\operatorname{Re}\frac{1}{\lambda+\psi}$$

for $\lambda>0$; see Sect. 4. The proof is complete.

For our final result we need to place a restriction on the hypergroup K.

7.10. **Theorem** (Sufficient condition for transience). *Let K be a strong hypergroup and let $(\mu_t)_{t>0}$ be a continuous convolution semigroup on K such that $\mu_0 = \varepsilon_e$, with corresponding strongly negative definite function ψ on $K^\wedge$. If* $\frac{1}{\psi}$ *is*

locally $\omega_{K^\wedge}$-integrable then $(\mu_t)_{t>0}$ is transient, and the potential measure κ of $(\mu_t)_{t>0}$ is given by

$$\kappa = \mathscr{F}_{K^\wedge}\left[\left(\frac{1}{\psi}\omega_{K^\wedge}\right)^-\right].$$

Proof. First we infer from Proposition 7.8(b) that the measure $\left(\frac{1}{\psi}\omega_{K^\wedge}\right)^-$ $\frac{1}{\bar{\psi}}\omega_{K^\wedge}$ is positive definite with Fourier transform

$$\sigma = \mathscr{F}_{K^\wedge}\left(\frac{1}{\bar{\psi}}\omega_{K^\wedge}\right) \in M^+(K).$$

Since K is assumed to be a strong hypergroup, we can apply Theorem 6.8 to the hypergroup $K^\wedge$ in place of K (see Remark 6.12) and to the measure $\mu := \frac{1}{\bar{\psi}}\omega_{K^\wedge} \in M_p(K^\wedge)$, to obtain for all $\phi \in C_c(K^\wedge)$ the formulae

$$\int_K |\bar{\mathscr{F}}_{K^\wedge}\phi(x)|^2\, d\sigma(x) = \int_{K^\wedge} \phi * \phi^\sim(\chi)\frac{1}{\bar{\psi}(\chi)}\, d\omega_{K^\wedge}(\chi)$$

and

$$\int_K |\bar{\mathscr{F}}_{K^\wedge}\phi(x)|^2\, d\rho_\lambda(x) = \int_{K^\wedge} \phi * \phi^\sim(\chi)\frac{1}{\lambda + \bar{\psi}(\chi)}\, d\omega_{K^\wedge}(\chi)$$

whenever $\lambda > 0$, where $(\rho_\lambda)_{\lambda>0}$ denotes the resolvent corresponding to $(\mu_t)_{t>0}$. Using Lebesgue's dominated convergence theorem we have

$$\lim_{\lambda\to 0}\int_{K^\wedge} \phi * \phi^\sim(\chi)\frac{1}{\lambda + \bar{\psi}(\chi)}\, d\omega_{K^\wedge}(\chi) = \int_{K^\wedge} \phi * \phi^\sim(\chi)\frac{1}{\bar{\psi}(\chi)}\, d\omega_{K^\wedge}(\chi),$$

that is,

$$\lim_{\lambda\to 0}\int_K |\bar{\mathscr{F}}_{K^\wedge}\phi(x)|^2\, d\rho_\lambda(x) = \int_K |\bar{\mathscr{F}}_{K^\wedge}\phi(x)|^2\, d\sigma(x) < \infty.$$

To every $f \in C_c^+(K)$ we have, by [9], Proposition 2, the existence of $\phi \in C_c(K^\wedge)$ satisfying $|\bar{\mathscr{F}}_{K^\wedge}\phi|^2 \geqq f$. This implies

$$\lim_{\lambda\to 0}\int_K f\, d\rho_\lambda \leqq \lim_{\lambda\to 0}\int_K |\bar{\mathscr{F}}_{K^\wedge}\phi(x)|^2\, d\rho_\lambda(x) < \infty$$

and therefore the transience of $(\mu_t)_{t>0}$.

Now let κ denote the potential measure of $(\mu_t)_{t>0}$. Then

$$\int_K |\bar{\mathscr{F}}_{K^\wedge}\phi(x)|^2\, d\kappa(x) = \lim_{\lambda\to 0}\int_K |\bar{\mathscr{F}}_{K^\wedge}\phi(x)|^2\, d\rho_\lambda(x)$$

for all $\phi \in C_c(K^\wedge)$ and hence, by the above formulae,

$$\begin{aligned}\int_K |\bar{\mathscr{F}}_{K^\wedge}\phi(x)|^2\, d\kappa(x) &= \int_K |\bar{\mathscr{F}}_{K^\wedge}\phi(x)|^2\, d\sigma(x)\\ &= \int_{K^\wedge} \phi * \phi^\sim(\chi)\frac{1}{\bar{\psi}(\chi)}\, d\omega_{K^\wedge}(\chi).\end{aligned}$$

Again applying Theorem 6.8 and Remark 6.12 yields $\kappa = \sigma$, and the proof is complete.

7.11. *Remark.* Let $\mu \in M^+(K)$ with $\mu(K) \leqq 1$ such that $\sum_{n \geqq 0} \mu^n < \infty$ in the sense of $\mathscr{T}_v$. Then $\operatorname{Re} \frac{1}{1-\hat{\mu}}$ is locally $\omega_{K^\wedge}$-integrable. This statement follows from Theorem 7.9 since the strongly negative definite function corresponding to the Poisson semigroup $(e^{-t} \exp(t\mu))_{t>0}$ is just $1 - \hat{\mu}$.

7.12. *Discussion.* If K is a locally compact abelian group, Theorem 7.10 can be strengthened by replacing the local $\omega_{K^\wedge}$-integrability of $\frac{1}{\psi}$ by that of $\operatorname{Re} \frac{1}{\psi}$. In this case Theorems 7.9 and 7.10 yield the *criterion* that a continuous convolution semigroup $(\mu_t)_{t>0}$ on K with $\mu_0 = \varepsilon_e$ is transient if and only if $\operatorname{Re} \frac{1}{\psi}$ is locally $\omega_{K^\wedge}$-integrable. Even in the group case the strong form of Theorem 7.10 has not yet been proved purely analytically and, of course, the probabilistic method applied (in [12]) is far from being accessible within the framework of hypergroups.

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